Benchmarking Long-Context Genomic Language Models for Synteny-Aware Antimicrobial Resistance Prediction

Background and Problem: Antimicrobial resistance (AMR) characterization relies on standard homology tools (e.g., BLAST or HMMER) that evaluate genes in isolation, ignoring regulatory syntenic context, pseudogene false positives, and flanking mobile genetic elements (MGEs) [1, 2]. Resolving complex genomic syntax requires modeling long-range neighborhoods [2, 3]. While genome language models (GLMs) capture complex biological grammar, localized baselines are constrained by quadratic Transformer self-attention ($\mathcal{O}(N^2)$), rendering wide-context sequence modeling computationally prohibitive [4]. Sub-quadratic architectures provide the scalability needed for synteny-aware AMR modeling.

Objectives and Hypotheses:

- **Objective 1 (Syntenic Mobilization):** Benchmark sub-quadratic models (1–32 kb) for detecting distal mobilization machinery (transposases, integrases, inverted repeats).
- **Objective 2 (Functional Integrity):** Validate single-nucleotide sensitivity to open reading frame (ORF) frameshifts and –10/–35 promoter motifs.
- **Hypothesis (a):** Sub-quadratic scaling (to 32 kb) with reverse-complement (RC) equivariance outperforms homology and localized baselines in identifying mobilization context and inverted repeats.
- **Hypothesis (b):** Single-nucleotide resolution modeling improves classification over standard homology and localized baselines by resolving pseudogenes and inactive promoters.

Methodology:

- **Dataset Curation & Leakage Mitigation:** Use assemblies with verified AST (antimicrobial susceptibility testing) metadata (NCBI [5] / BV-BRC [6]) annotated via CARD [7]; cluster at 60–80% identity (MMseqs2 [8]) to prevent homology leakage. Label Active Positives (intact ORF, functional promoter, adjacent MGE) alongside three negative controls: Type A (non-mobile homologs), Type B (frameshifted pseudogenes), and Type C (non-functional promoter variants).
- **Model Fine-Tuning & Benchmarking:** Fine-tune sub-quadratic models (HyenaDNA [4]; RC-Caduceus [9]) across 1, 8, and 32 kb using parameter-efficient adapters, evaluating against a 1 kb localized baseline (DNABERT-2 [10]) and homology tools (AMRFinderPlus [1]). A dual-output head will jointly classify gene functionality and mobilization status, with loss penalties for class imbalance.
- **Out-of-Distribution Generalization:** Train on *Enterobacteriaceae* (50–54% GC) and evaluate out-of-distribution on *Acinetobacter baumannii* (39% GC) and *Pseudomonas aeruginosa* (66% GC) to test GC-bias robustness.

Model Validation & Interpretability: Performance will be evaluated via Precision-Recall AUC (PR-AUC) and Matthews Correlation Coefficient (MCC). Integrated Gradients (IG) [11] and In Silico Mutagenesis (ISM) [12] will confirm single-nucleotide attributions to –10/–35 promoter motifs and validate distal epistatic interactions across 32 kb windows. Fast Fourier Transform (FFT) analysis on convolutional filters will detect periodic biophysical motifs (e.g., the 10.5 bp DNA helical pitch) [13].

Project Feasibility: Supported by HPC resources (NVIDIA GPU clusters) from the Digital Research Alliance of Canada, the 24-month project progresses from data curation (Months 1–6) to model fine-tuning & benchmarking (Months 7–15) and interpretability analysis & tool deployment (Months 16–24).

Relevance & Potential Contributions: This project benchmarks sub-quadratic, RC-equivariant models for synteny-aware AMR sequence modeling. Open-sourcing our curated benchmark dataset, fine-tuned checkpoints, and validation pipelines provides reproducible tools to reduce pseudogene false positives and resolve horizontal gene transfer in bacterial genomics.

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References

[1] M. Feldgarden et al. "Validating the AMRFinder Tool and Resistance Gene Database by Using Antimicrobial Resistance Genotype-Phenotype Correlations in a Collection of Isolates". eng. In: *Antimicrobial agents and chemotherapy* 63.11 (2019).

[2] B. Jia. "Advancing Antimicrobial Resistance (AMR) Surveillance: Characterizing AMR Mobility and Improving AMR Prediction from Metagenomics Data". en. Ph.D. dissertation. Burnaby, BC, Canada: Simon Fraser University, 2022.

[3] L. Shu et al. "A comprehensive survey of genome language models in bioinformatics". en. In: *Briefings in Bioinformatics* 27.1 (Jan. 2026), bbaf724.

[4] E. Nguyen et al. *HyenaDNA: Long-Range Genomic Sequence Modeling at Single Nucleotide Resolution*. Nov. 2023.

[5] E. W. Sayers et al. "Database resources of the National Center for Biotechnology Information in 2025". In: *Nucleic Acids Research* 53.D1 (Jan. 2025), pp. D20–D29.

[6] R. D. Olson et al. "Introducing the Bacterial and Viral Bioinformatics Resource Center (BV-BRC): a resource combining PATRIC, IRD and ViPR". eng. In: *Nucleic Acids Research* 51.D1 (Jan. 2023), pp. D678–D689.

[7] B. P. Alcock et al. "CARD 2020: antibiotic resistome surveillance with the comprehensive antibiotic resistance database". en. In: *Nucleic Acids Research* (Oct. 2019), gkz935.

[8] M. Steinegger and J. Söding. "MMseqs2 enables sensitive protein sequence searching for the analysis of massive data sets". eng. In: *Nature Biotechnology* 35.11 (Nov. 2017), pp. 1026–1028.

[9] Y. Schiff et al. *Caduceus: Bi-Directional Equivariant Long-Range DNA Sequence Modeling*. arXiv:2403.03234 [q-bio.GN]. June 2024.

[10] Z. Zhou et al. *DNABERT-2: Efficient Foundation Model and Benchmark For Multi-Species Genome*. Mar. 2024.

[11] M. Sundararajan, A. Taly, and Q. Yan. *Axiomatic Attribution for Deep Networks*. arXiv:1703.01365 [cs.LG]. June 2017.

[12] D. R. Kelley, J. Snoek, and J. L. Rinn. "Basset: learning the regulatory code of the accessible genome with deep convolutional neural networks". In: *Genome Research* 26.7 (July 2016), pp. 990–999.

[13] Ž. Avsec et al. "Base-resolution models of transcription-factor binding reveal soft motif syntax". en. In: *Nature Genetics* 53.3 (Mar. 2021), pp. 354–366.